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Where Minds Pollinate

Class 9
Chemistry

Chapter 13

Handwritten Notes

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Unit # 13

ORGANIC CHEMISTRY

~ Table ~

Alkanes	Alkenes	Alkynes
Bond:- Single	Bond:- double	Bond:- Triple
General formula:- C_nH_{2n+2}	General formula:- C_nH_{2n}	General Formula:- C_nH_{2n-2}
CH_4 (Methane)	—	—
C_2H_6 (Ethane)	C_2H_4 (Ethene)	C_2H_2 (Ethyne)
C_3H_8 (Propane)	C_3H_6 (Propene)	C_3H_4 (Propyne)
C_4H_{10} (Butane)	C_4H_8 (Butene)	C_4H_6 (Butyne)
C_5H_{12} (Pentane)	C_5H_{10} (Pentene)	C_5H_8 (Pentyne)
C_6H_{14} (Hexane)	C_6H_{12} (Hexene)	C_6H_{10} (Hexyne)
C_7H_{16} (Heptane)	C_7H_{14} (Hepene)	C_7H_{12} (Heptyne)
C_8H_{18} (Octane)	C_8H_{16} (octene)	C_8H_{14} (Octyne)
C_9H_{20} (Nonane)	C_9H_{18} (Nonene)	C_9H_{16} (Nonyne)
$C_{10}H_{22}$ (Decane)	$C_{10}H_{20}$ (Decene)	$C_{10}H_{18}$ (Decyne)

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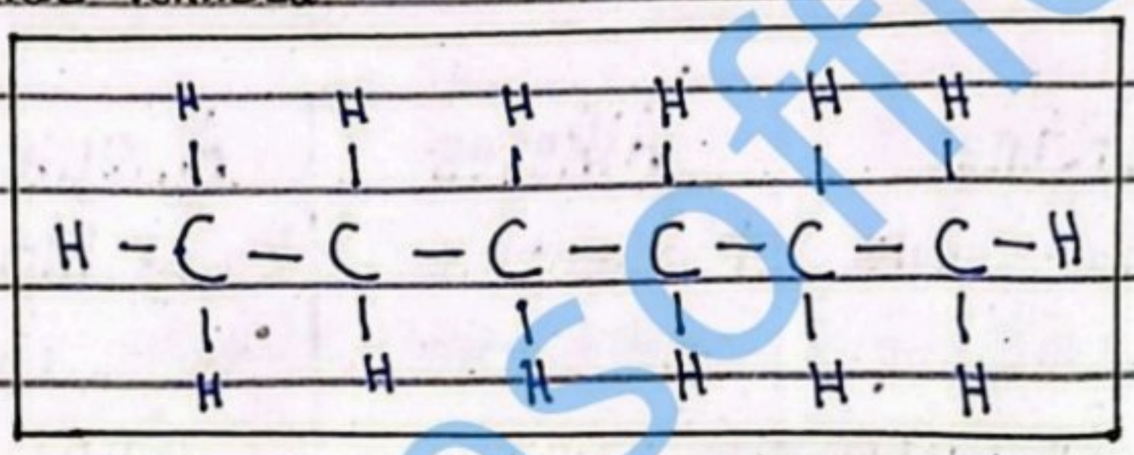
Question no 1:- Write the molecular formula, Structural formula and Condensed Structural formula of Hexane.

Answer:-

Molecular formula:-

C_6H_{14}

Structural formula:-



Condensed Structural formula:-

$CH_3-CH_2-CH_2-CH_2-CH_2-CH_3$

Question no 2:- Write the molecular formula, Structural formula and Condenced Structural formula of the following

a:- Hexene

b:- Pentyne

c:- Butane.

Answer:-

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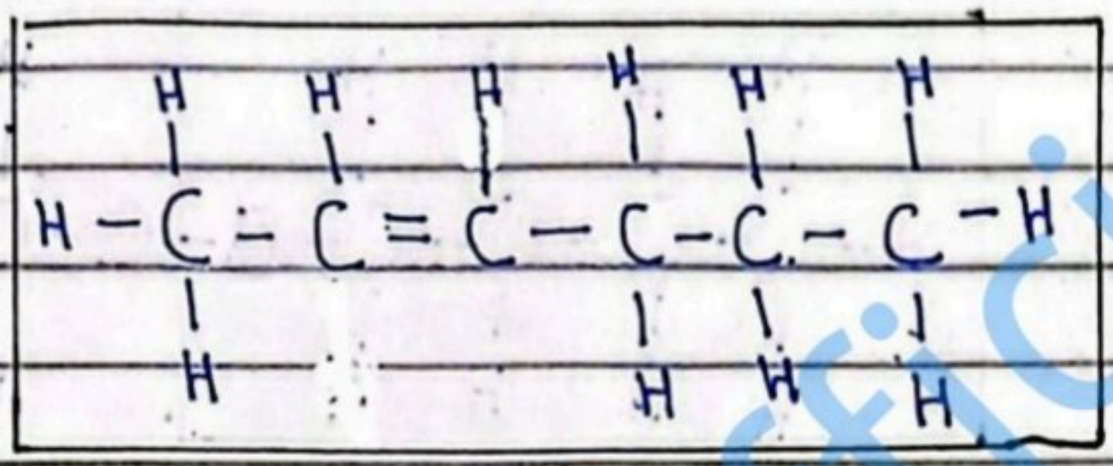
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a)

Molecular formula:-

C_6H_{12}

Structural formula:-



Condenced Structural formula:-

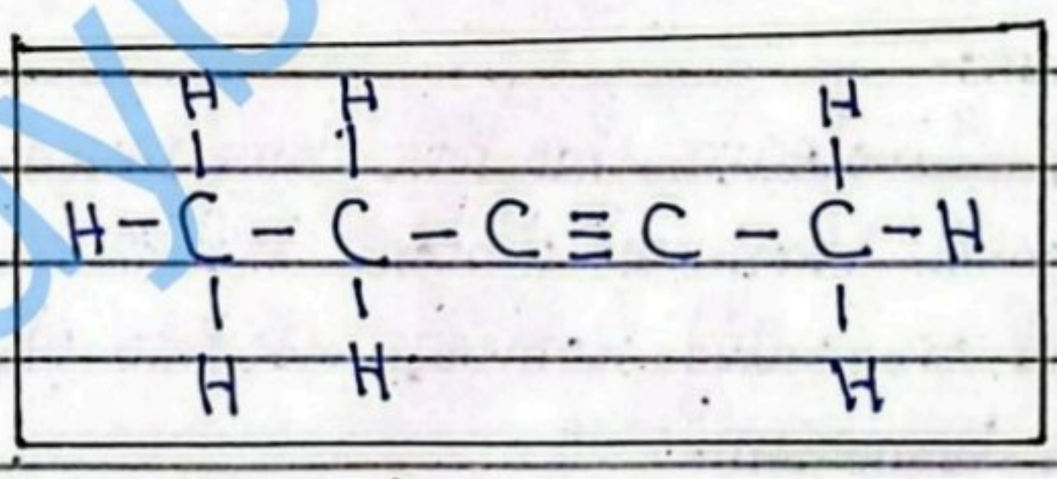
$CH_3-CH=CH-CH_2-CH_2-CH_3$

b)

Molecular formula:-

C_5H_8

Structural formula:-



Condenced Structural formula:-

$CH_3-CH_2-C\equiv C-CH_3$

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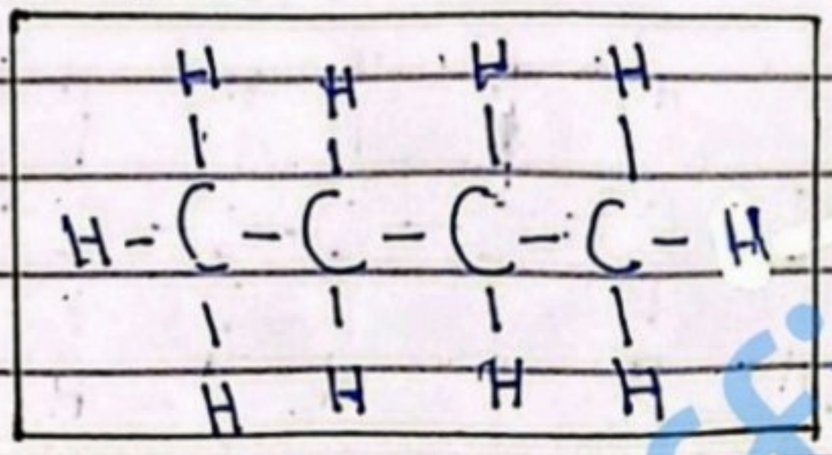
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c)

§ Molecular formula:-

C_4H_{10}

Structural formula:-



Condenced Structural formula:-

$CH_3-CH_2-CH_2-CH_2-CH_3$

Question No 3:- Define isomerism and explain it with examples?

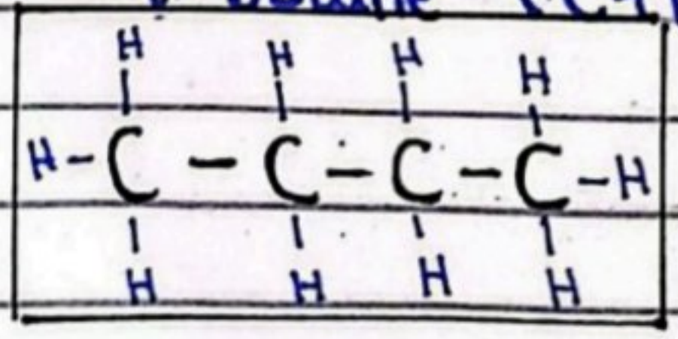
Answers:-

Isomerism:-

"Molecules that have Same molecular formula but different arrangement of atoms in their molecule Structure are called isomers, and this phenomia is known as isomerism."

Example:-

i Butane (C_4H_{10})



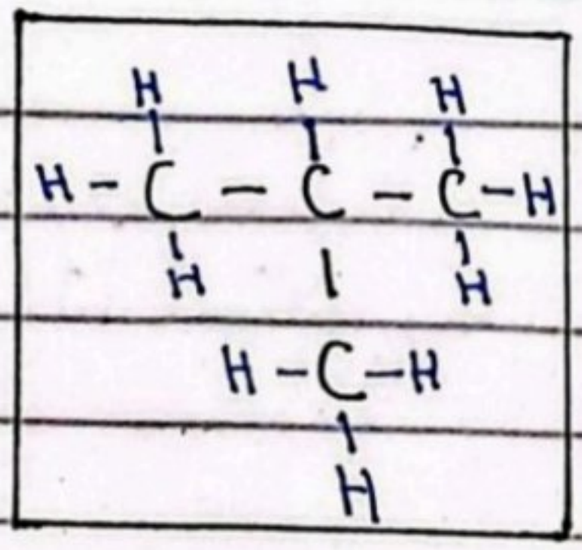
"n-Butane"

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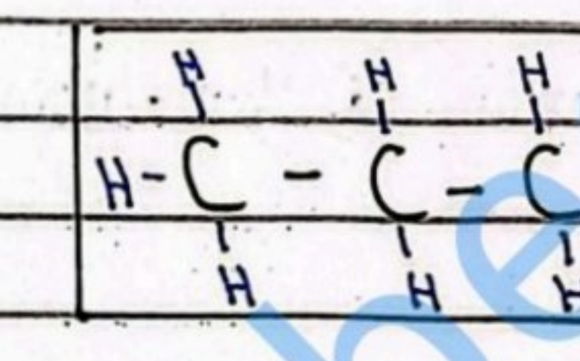
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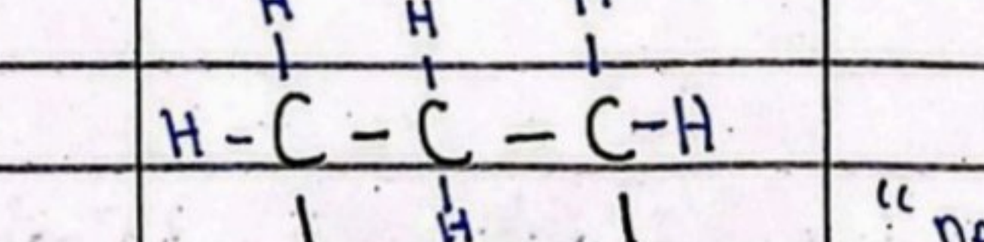


"iso-Butane"

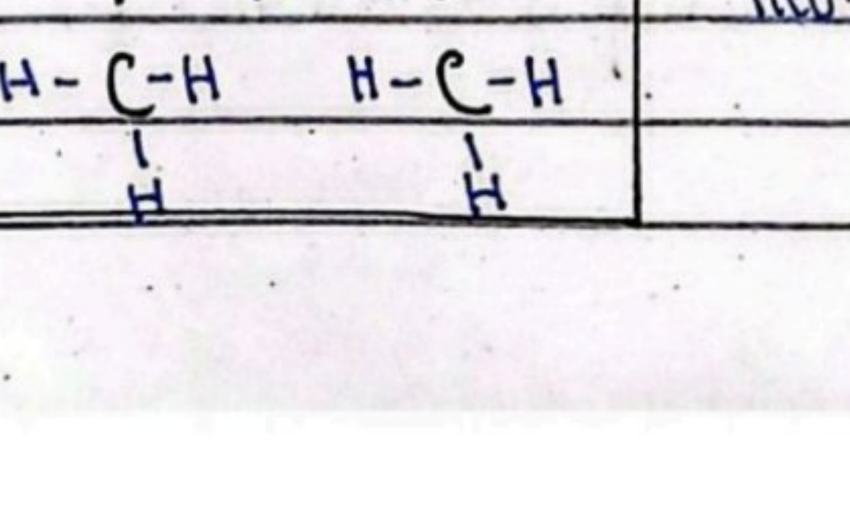


"neo-Butane"

ii Pentane (C_5H_{12})



"n-Pentane"



"iso-Pentane"

"neo-Pentone"

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Concept Assessement No 13.1

Question:-
Write the molecular formulae of the following Compounds using general formulae

i) Alkane Containing i) 4 C-atom ii) 6 H-atom
Sol:-
General formula:- C_nH_{2n+2}
i) C_4H_{10}
ii) C_6H_{14}

ii) Alkene Containing i) 3 C-atom ii) 4-C-atom
Sol:-
General formula:- C_nH_{2n}
i) C_3H_6
ii) C_4H_8

iii) Alkyne Containing i) 3 C-atom ii) 4 H-atom
Sol:-
General formula:- C_nH_{2n-2}
i) C_3H_4
ii) C_4H_6

iv) Alcohol Containing i) 2 C-atom ii) 3 H-atom
Sol:-

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General formula:- $C_nH_{2n+1}OH$

i) C_2H_5OH

ii) C_3H_7OH

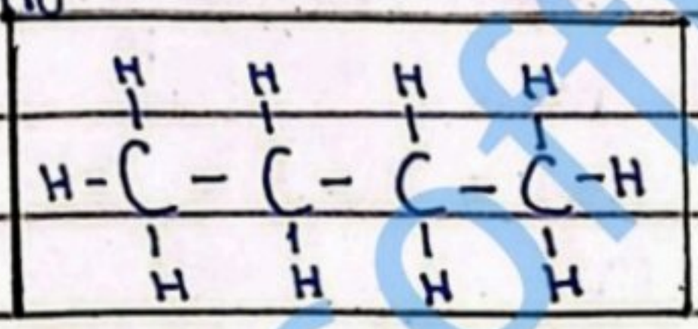
Concept Assessement No 13.2

Write, Molecular, Structural and Condenced formula

a) Butane:-

Molecula:- C_4H_{10}

Structural:-

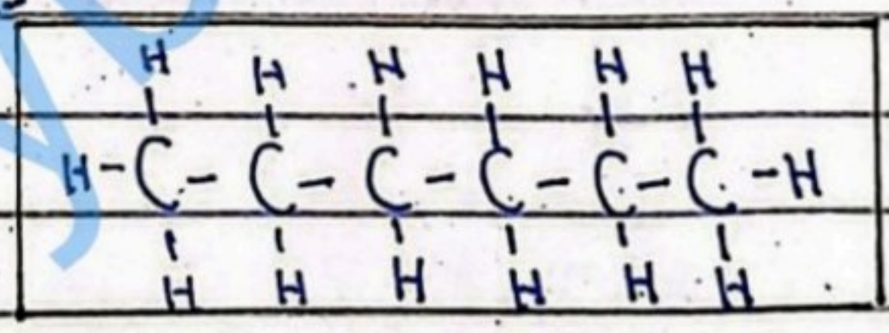


Condenced:- $CH_3-CH_2-CH_2-CH_3$

b) Hexane:-

Molecular:- C_6H_{12}

Structural:-

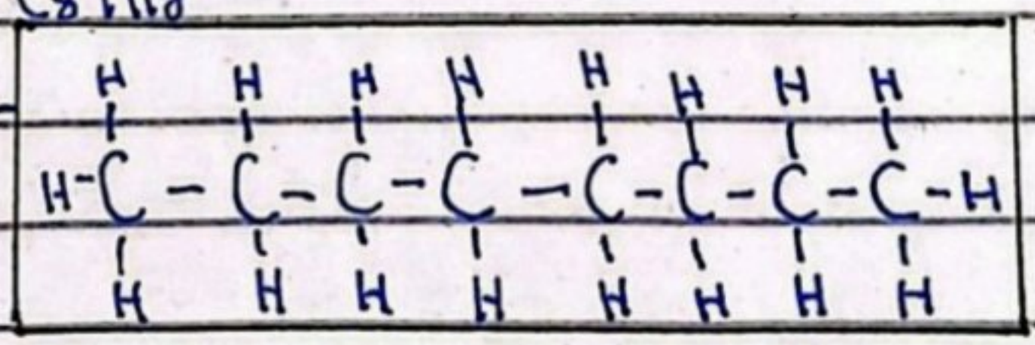


Condenced:- $CH_3-CH_2-CH_2-CH_2-CH_2-CH_3$

b) Octane:-

Molecular:- C_8H_{18}

Structural:-



Condenced:- $CH_3-CH_2-CH_2-CH_2-CH_2-CH_2-CH_2-CH_3$

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Concept Assessement No 13.3

Question :-

Choose Saturated and unsaturated Compound of the following.

i) $CH_3 - C \equiv CH$

Unsaturated Compound, Containing 'triple bond'.

ii) $CH_3 - CH_2 - CH_3$

Saturated Compound, Containing 'Single bond'.

iii) $CH_3 - CH = CH_2$

Unsaturated Compound, Containing 'double bond'.

iv) $CH_2 = CH - CH = CH_2$

Unsaturated Compound, Containing 'double bond'.

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Question No 4:- What is homologus Series? explain.

Answer:-

Homologous Series:-

" A Series of organic Compounds in which any two molecules differ by $-CH_2$ group is called homologus Series"

Example:-

Alksones, Alksenes and Alksynes

Consider Alksanes:-

CH_3-H , CH_3-CH_3 , $CH_3-CH_2-CH_3$,

Methane Ethane propane

$CH_3-CH_2-CH_2-CH_3$

Butane.

Similary Alcohal also from homdogus Serjies:-

CH_3-OH , CH_3-CH_2-OH , $CH_3-CH_2-CH_2$

Methanal Ethanol Propanal

$-OH$.

Aldeydes, Ketones, Carboxylic acids etc from homologus Sevies.

Question no 5:- What are saturated and uncaturated hydrocarbons ?

Answer:-

Hydrocarbon "Saturated":-

"Hydrocarbon Canhaing a Single

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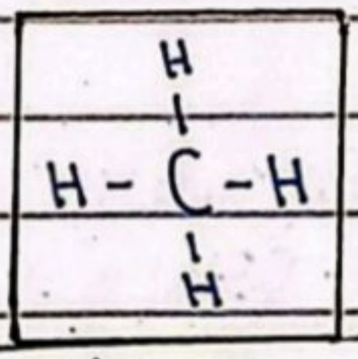
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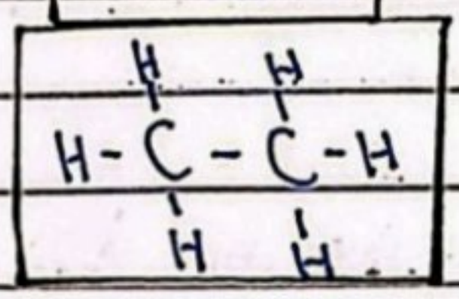
bond (—) between 2-Carbon-atom are called Saturated hydrocarbon"

Example:-

i) Methane:-



ii) Ethane:-

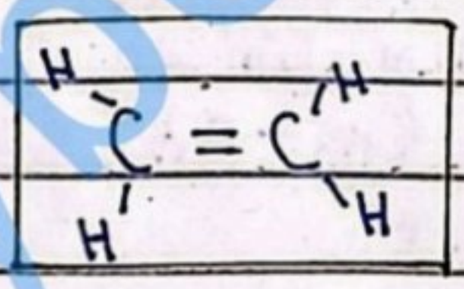


Unsaturated hydrocarbon:-

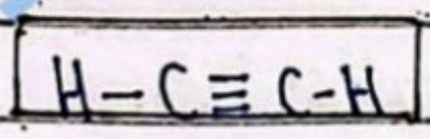
"Hydrocarbon Containing double (=) or triple bond (≡) between 2-Carbon-atom is called unsaturated hydrocarbon.

Example:-

i) Ethene:-



ii) Ethyne:-



Question No 7:- What are the general characteristics of a homologus Series?

Answer:- General Characteristics of homologus Series:-

i) The family members of a homologus Serdies have the Same functional group.

ii) The family member have Same general formula.

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iii) The adjacent family members differs by a -CH2- unit.

iv) The family member display a trend in their physical properties.

v) The family members possess similar chemical properties.

Question No 8:- Define catenation and explain with the help of an example.

Answer:-

Catenation:-

Definition:-

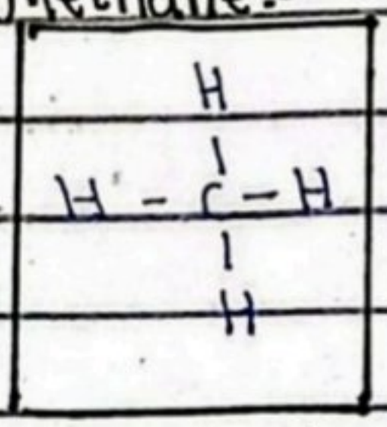
"The Self-linking ability of carbon is called catenation.

• Carbon has four bonding electrons in it's valence shell. Carbon, therefore forms four bonds with other atoms.

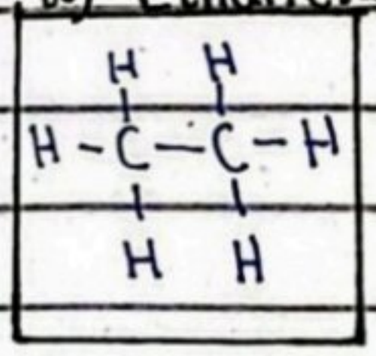
• The chemical diversity of organic Compounds arises from Carbon's ability to bond to each other to form long chain, branched chain and ring.

• No other element can complete with carbon in this regard Silicon and few other element can form chains but only short one. Carbon chain may Contain thousands of carbon atom.

i) Methane:-



ii) Ethane:-



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Question No 9:- Write a detail note on the classification of organic Compounds ~~exemple~~ explain with example.

Answer:- These are many millions of organic Compounds. It is not easy to study each Compounds Separetely. They are divided into groups and Sub-group. They are divided into two main group.

1- Open chain or Acyclic Compounds.
2- Closed chain or cyclic Compounds.

Open chain Compounds

Open chain Compounds Contain an open chain of Carbon atom. In these Compounds Carbon atoms are linked in Linear pattern.

Example:-

- $CH_3 - CH_2 - CH_3$ (Propane)
- $CH_3 - CH_2 - CH_2 - CH_3$ (Butane)
- $CH_3 - CH_2 - CH_2 - CH_3$ (iso-Pentane)

CH_3

Open chain Compounds may be either straight chain or branched chain.

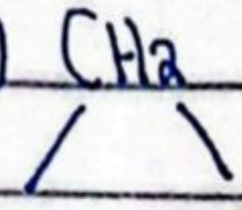
Closed chain Compounds

Organic Compounds which Contain rings of atoms are called closed chain or cyclic Compounds.

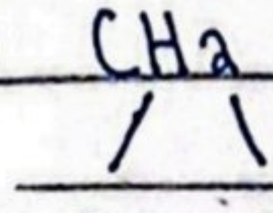
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Example:-

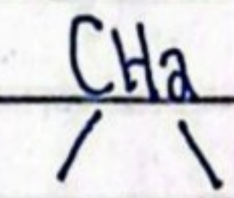
i) $CH_3 - CH_2 - CH_3$ = Propane

i) 

Cyclo-propane.

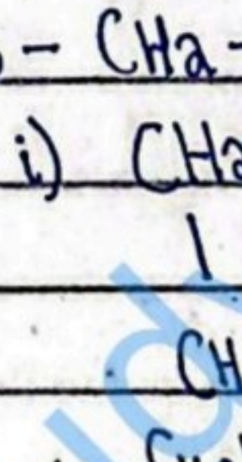
ii) 

Cyclo-propene.

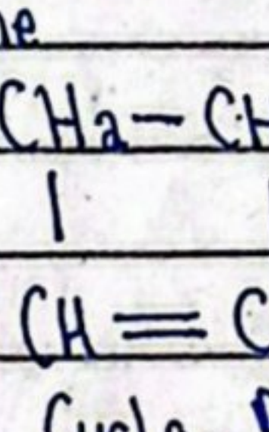
iii) 

Cyclo-propyne.

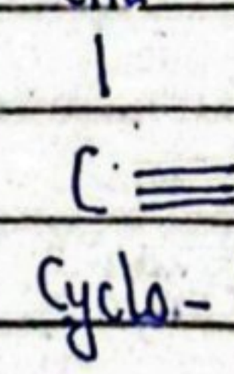
ii) $CH_3 - CH_2 - CH_2 - CH_3$ = Butane

i) 

Cyclo-Butane.

ii) 

Cyclo-Butene.

iii) 

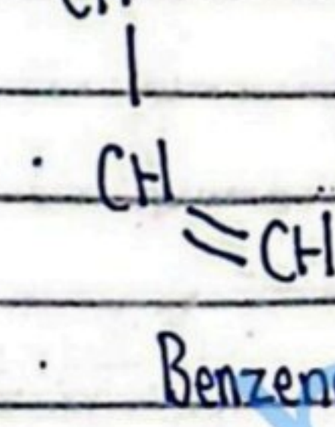
Cyclo-Butyne.

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iii) $CH_3 - CH_2 - CH_2 - CH_2 - CH_3$

i) 

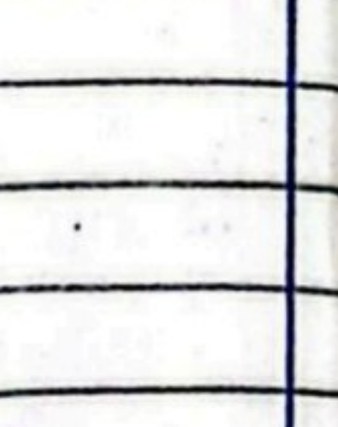
$CH_2 - CH_2$

Cyclo-pentane

ii)

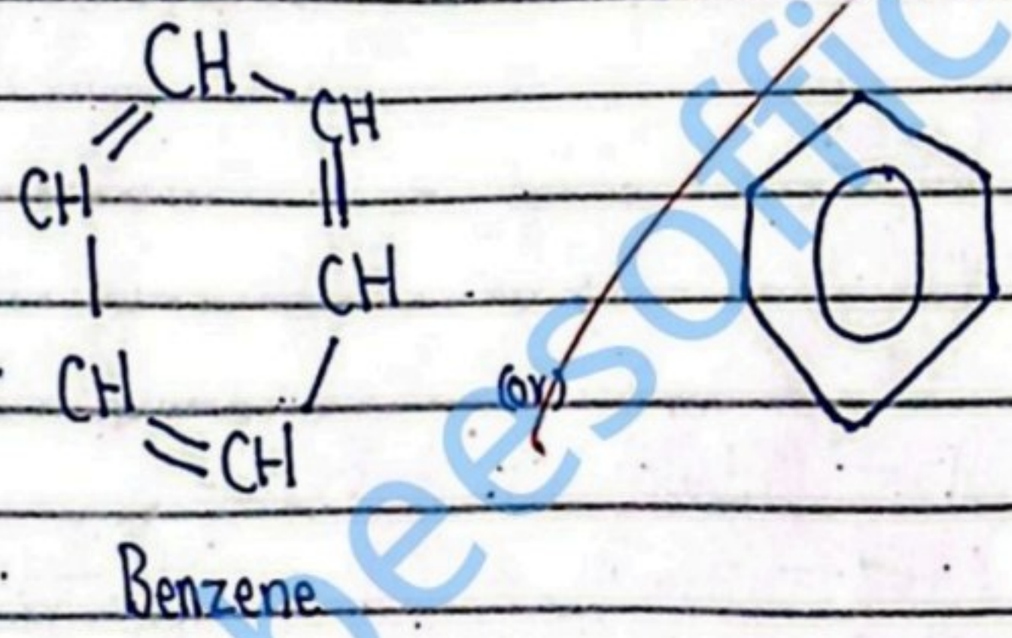
$CH = CH$

Cyclo-Pentene

iii) 

$C \equiv C$

Cyclo-Pentyne

iv) 

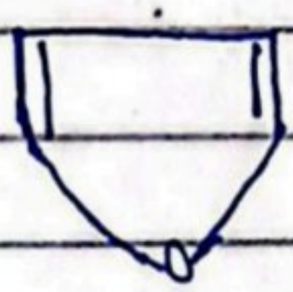
Benzene

Cyclic Compounds which Contain rings of carbon atoms are called homocyclic or carboxcyclic Compounds. Which of the above cyclic Compounds are carbocyclic? Cyclic Compounds that Contain one or more atoms in the ring are called heterocyclic Compounds.

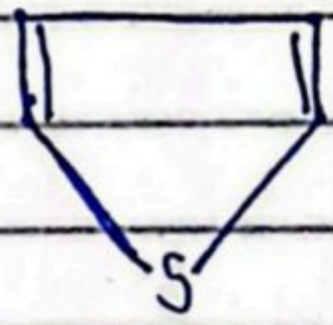
Example::

i) 

Pyridine

ii) 

Furan

iii) 

Thiophene

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Question No 10:- What are alkyl radicals? Write it's general formula and explain with examples.

Answer:-

Alkyl Radicals :-

Alksyl Radicals are formed by removing one Hydrogen atom from alksanes.

General Formula:-

It's general formula is "C_nH_{2n+1}"

It is represented by "R"

Example:-

Alkanes	Alkyl Radicals
i CH_4 Methane	CH_3- Methyl
ii CH_3CH_3 Ethane	CH_3-CH_2- Ethyl
iii $CH_3-CH_2-CH_3$ Propane	$CH_3-CH_2-CH_2-$ Propyl

Question No 11 :- Explain functional group, explain different groups with examples.

Answer:-

Functional Group:-

" An atom or goup of atom that give the family of orgenic Compounds it's characteristics, physical and chemical properties is known as functional group.

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Each functional group defines of organic Compounds. There are millions of organic Compounds but they have few functional groups. Each functional group determines the characteristic properties of homologus Series.

Following are Some important functional group:-

- **HaloAlkanes or Alkyl hallides:-**

Haloalkanes are characterized by the presence of halogen atom with alkyl radical.

General formula:-

$(R-X)$

R(alkyl radicals) ; X (halogen atom Such as F, Cl, Br, I)

Example:- • CH_3-Cl

'Methyl Chloride' or 'chlora methane'

• C_2H_5-Cl

'Ethyl chloride' or 'chloro ethane'

- **Alcohol:-**

Alcohol characterized by the presence of hydroxyl group (-OH) attached to hydro carbon chain.

General formula:-

$(R-OH)$

Examples :- • CH_3-OH

'Methanol' or 'Methyl alcohol'

• C_2H_5

'Ethanol' or 'Ethyl alcohol'

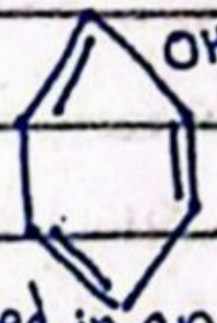
- **Phenol:-**

When an (-OH) group is attached to benzene ring the Compound is called phenol.

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Example:-

C_6H_5-OH or 

Phenol was the first anticeptic used in an Operation theatre.

- **Ether:-**

Organic Compounds that have two alkyl grasp attached to the Same oxygen atom is called ether.

General formula:-

$R-O-R'$

Example:-

- CH_3-O-CH_3 (Dimethyl ether)
- $C_2H_5-O-CH_3$ (ethyl methyl ether)

- **Aldehyde and ketone:-**

Aldehyde and ketone contain carbonyl group $-\overset{O}{\overset{\|}{C}}-$ or $-CO-$ An aldehyde has at least one hydrogen atom while ketone has 2 hydrocarbon groups (alkyl) bonded to the carbonyl carbon atom.

Example:- Aldehyde

- $H-\overset{O}{\overset{\|}{C}}-H$ or $HCOH$

Formaldehyde or methanol

- $H-\overset{O}{\overset{\|}{C}}-CH_3$ or $HCOCH_3$

Acetaldehyde or Ethanol

Example:- ketone

$CH_3-\overset{O}{\overset{\|}{C}}-CH_3$ or CH_3COCH_3

Propanone or (Acetone)

$CH_3-\overset{O}{\overset{\|}{C}}-C_2H_5$ or $CH_3OCC_2H_5$

Butanone

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- **Carboxylic Acid:-**
It contain carboxyl group $\left(-\overset{O}{\overset{\|}{C}}-OH\right)$ or $(-COOH)$

Example:-

- $H-\overset{O}{\overset{\|}{C}}-OH$ or $HCOOH$

  Formic acid or Methanoic acid

- $CH_3-\overset{O}{\overset{\|}{C}}-OH$ or CH_3-COOH

  Acetic acid or Ethanoic acid

- **Esters:-**
Their general formula is $R-\overset{O}{\overset{\|}{C}}-O-R'$

Example:-

- $CH_3-\overset{O}{\overset{\|}{C}}-O-CH_3$ or CH_3COOCH_3

  methyl acebate or methyl Ethanate

- $CH_3-\overset{O}{\overset{\|}{C}}-O-C_2H_5$ or $CH_3COOC_2H_5$

  Ethyl acetate or Ethyl Ethanate:

- **Amides:-**
Their general formula is $R-\overset{O}{\overset{\|}{C}}-NH_2$

Example:-

- $CH_3-\overset{O}{\overset{\|}{C}}-NH_2$ or CH_3CONH_2

  Ethanamide

- **Amines:-**
Functional group of amine is $-NH_2$

Example:-

- CH_3-NH_3

  methyl amine

- $C_2H_5-NH_2$

  Ethyl amine

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Chp 13 :

-:Example 13.2:-

Differentiate between organic Compounds on the basic of functional Group.

-:Solution:-

i) $CH_3CH_2OCH_2CH_3$

Ans:- Ether (Di-ethyl ether)

ii) C_6H_5-OH

Ans:- Phenol

iii) CH_3-OH

Ans:- Alcohol (methanol)

Concept Assessement Exercise 13.5

Classify the following Compounds as alcohal, ether, phenol.

a) $CH_3CH_2OCH_2CH_3$

Ans:- Ether (Di-ethyl ether)

b) $CH_3CH_2CH_2OH$

Ans:- Alcohol (Propanol or proply alcohal)

c) C_6H_5OH

Ans:- Phenol

d) C_2H_5OH

Ans:- Alcohol (Ethanol)

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-: Example 13.3 :-

Identify the following Compounds as an aldehyde or a Ketone or a Carboxlyic acid.

-: Solution :-

i) CH_3COCH_3 or $CH_3-\overset{O}{\overset{\|}{C}}-CH_3$

Answer:- "Ketone"

ii) CH_3CH_2CHO or $CH_3-CH_2-\overset{O}{\overset{\|}{C}}-H$

Answer:- Aldehyde

iii) CH_3COOH or $CH_3-\overset{O}{\overset{\|}{C}}-OH$

Answer:- "Carboxlic acid"

Concept Assessement [Exercise 13.6]

Identify an aldehyde, ketone, or a carboxlyic acid.

a) $CH_3COCH_2CH_3$ or $CH_3-\overset{R}{C}-CH_2-CH_3$

Answer:- Ketone [Butanone]

b) $CH_3-CH_2-\overset{R}{C}-H$

Answer Aldehyde [Propanal]

c) $CH_3-CH_2-\overset{R}{C}-OH$

Answer:- Carboxylic acid [Propanoic acid]

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Exercise No 13:

Question no 3: Identify the following on the basis of functional groups they contain and encircle the functional group.

i) $CH_3-CH=CH_2$

Ans:- Alkene.

ii) CH_3-CH_2-COOH

Ans:- Carboxylic acid.

iii) $CH_3-\overset{\overset{O}{\|}}{C}-OH$

Ans:- Carboxylic acid.

iv) $CH_3-\overset{\overset{O}{\|}}{C}-CH_3$

Ans:- Carbonyl group.

v) $CH_3-\overset{\overset{O}{\|}}{C}-OCH_3$

Ans:- Ester.

vi) $H-\overset{\overset{O}{\|}}{C}-CH_3$

Ans:- Carbonyl group.

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Question no 4:- What is the name of alkane having four Carbon atom in chain?

Answer:-

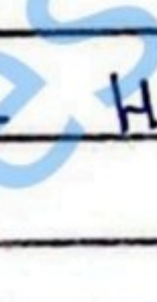
The name of alkane having four atoms in chain is Butane.

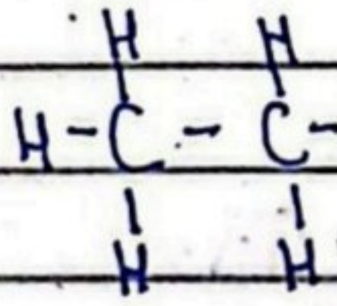
$CH_3 - CH_2 - CH_2 - CH_3$

Question no 5:- Give the structural formula of two Simple Simple alkanes and one alkyne.

Answer:-

i) Alkanes:-

Methane:- 

Ethane:- 

ii) Alkyne:-

Ethyne:- C_2H_2 $H-C\equiv C-H$

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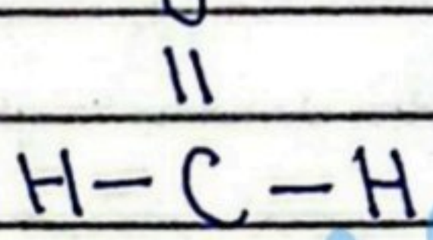
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Question no 7:- Identify the type of following Compound as ~~alk~~ alcohol, ketone, Aldehyde

Answer:-

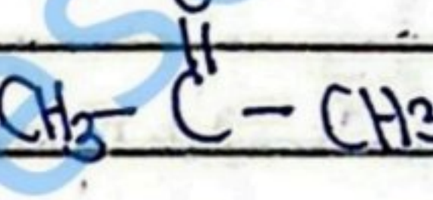
a)

HCHO or



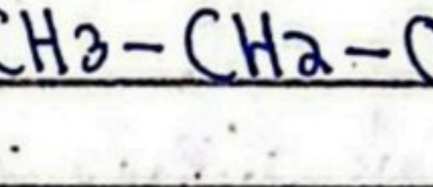
"Aldehyde"

b) CH_3COCH_3 or



"ketone"

c) CH_3CH_2OH or



"Alcohol"

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5th Oct - 25 H.W Monday

CHP # 14

Question No 180 ~
Write the methods of preparations of Alkanes?
Answer:-
Preparation of alkanes:-
We can perpare alkanes by using different methods Some of them are written and dissucs below:-
1. Hydrogenation of alkenes and alkynes:-
- Addition of hydrogen molecule across Carbon Carbon multiple bond is called hydrogenation.
- It take place in the presence of nickel at the temperture of 200-300°C at very high pressure.
- Hydrogenation can also done in the presence of Pt or Pd at room temperture.

Example:-

$$CH \equiv CH + H_2 \xrightarrow{Ni,200\text{-}300^\circ C} CH_2 = CH_2$$
Ethyne Ethene

$$CH_2 = CH_2 \xrightarrow{Ni,300\text{-}300^\circ C} CH_3 - CH_3$$
Ethene Ethane

2. Reduction of alkyl halides:-

- When an alkyl halide is treated with Zn in presence of an aqueous acid, an alkanes is produce.
- Usally aqueous Solution of HCL and CH_3COOH is used.

Example:-

$$CH_3-CL + 2[H] \xrightarrow{Zn+H_2O(aq)} CH_4 + HCL$$

- Zn reacts with aqueous acid to Liberate atomic Hydrogen
- called nascent hydrogen.
- Nacent hydrogen reduces alkyl halide.
- Addition of nacent hydrogen is called reduction.

3. Cracking of Large hydrocarbon:-

- A large hydrocarbon (alkane) is heated at very high temperture Large hydrocarbon are broken into Small hydrocarbon and this process is called thermal cracking.
- Temperture between 450 to 750°C at very high pressure.

Example:-

When decane is heated at high temperture and pressure, it break into octane and ethene.

$$C_{10}H_{22} \xrightarrow{450-750^\circ C} C_8H_{18} + C_2H_4$$

Decane Octane Ethene.

Toppersnotes

Question no 2:- Write the properties of alkanes?.

Answer:-

Properties of allsanes:-

Allcane molecule are essentially non-polar. They are less dence than water and don't dissolved in it. They don't react with most ionic Compounds. The lack of reactivity malses them useful Solvents.

For example:-

hexane is used to exact vegetable oils from Corn, Soybeans Cotton Seeds etc.

1) Alkanes Containing up to four Carbon atom is called Colourless and odourless gases.

2) Alkanes Containing 5 to 7 carbon atom ared orderless & Colourless liqpids.

3) Alkanes that are Higher are orderless and Cobuless Solids.

Halogenation:-

Allsanes readily react with halogens in Sunlight. A chemical reaction that talce place in the presence of Sunlight is called photochemical reaction. The reaction of an alkane and a halogen is a Substitution reaction. In this reaction a halogen atom substitutes for one or more of the hydrogen atom of a allcanes.

For example:-

The reaction of methane and chlorine in diffused Sunlight occur as follows:-

$$CH_4 + Cl_2 \xrightarrow{\text{diffused sunlight}} CH_3Cl + HCl$$

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In direct Sunlight the reaction of methane and chlorine in ~~diffused~~ direct Sunlight occur as follows:-

$$CH_4 + 2Cl_2 \xrightarrow{\text{direct Sunlight}} 4HCl + C$$

Combustion:-

A reaction of a Substance with oxygen or air Cause the rapid release of heat and the appearance of a flame is called Combustion.

Complete Combustion of an alkane produces CO_2, H_2O and heat. Most of alkanes burn with blue flame

For example:-

following reaction occurs when natural gas is burned:-

$$CH_4 + 2O_2 \longrightarrow CO_2 + 2H_2O + \text{heat}$$

In Complete Combustion occur in presence of Limited Supply of Oxygen. Incomplete Combustion of methane gives CO, C, and H_2O.

For example:-

$$3CH_4 + 4O_2 \longrightarrow 2CO + C + 6H_2O$$

Question No 3:- Write the uses of alkanes as fuel?

Answer:-

The Lighter alkanes are widely used as fuel because:-

1) There Combustion can be Control.
2) They produce large amount of heat per gram.
3) They are cheap and readily available.

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Where Minds Pollinate

Class 9
Chemistry

Chapter 13

Handwritten Notes (2)

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Unit :- 13

Date 8 June 20 26
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Organic Chemistry

- **Definition:-**
  Study of carbon compounds. (organic compounds)

- Carbon has ability to link with itself n other atoms to form a large no. of compounds.

Catenation
self-linking ability of C

- **Organic Compounds :-**
  → Compound that contain Carbon
  → also defined as hydrocarbons and their derivatives.
  Compounds contain hydrogen nd carbon only

- **Hydrocarbons :-**
  Hydrocarbons are divided into :-

Saturated

- Hydrocarbons whose C-C bonds are all single bonds.
- single bond = sigma bond
- also called Alkanes
- General formula = C_nH_{2n+2}
- Further molecule cannot be added - only substitution (replacement) happens.
- e.g :- Methane, Ethane, Propane.

Unsaturated

- Hydrocarbons that contain C-C multiple bonds (double or triple)
- Sigma + Pi bond
- Divided into 2 types.

Alkenes

- Contain atleast 1 =
- General formula C_nH_{2n}
- e.g :- Ethene
- Meth doesn't exist in alkenes.

Alkynes

- Contain atleast 1 ≡
- General formula C_nH
- e.g :- Ethyne
- Meth doesn't exist in alkynes.

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• Bond :-

	Sigma Bond	Pi Bond
Overlap	head to head	side by side
Strength	strong	Weak
Breaking	Hard to break	Easy to break
Present in	Single, double n, triple bond	Only in double n triple bond.

Single bond = 1 sigma bond
Double bond = 1 sigma + 1 pi bond
Triple bond = 1 sigma + 2 pi bond

• Substitution rxn :-

→ Rxn of saturated hydrocarbons
→ One atom or group replaces another atom in molecule.
→ further molecule cannot be added, only replacement occurs
→ Eg :- $CH_4 + Cl_2 \longrightarrow CH_3Cl + HCl$

• Addition rxn :-

→ Rxn of unsaturated Hcarbons
→ A molecule is added across the double or triple bond.
→ Pi bond breaks (bcz Pi is weak n easy to break)
→ Example :- $CH_2 = CH_2 + H_2 \longrightarrow CH_3CH_3$

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• Elimination Rxn :-

Opposite of addition : a molecule is removed.

• Condensation Rxn :-

two molecules join together with loss of small molecule (water).

* Why Meth doesn't exist in unsaturated hydrocarbons?

Meth doesn't exist in unsaturated bcz 1 C cannot form a double or triple bond with itself.

* Homologous Series

Alkanes	C_nH_{2n+2}
Alkenes	C_nH_{2n}
Alkynes	C_nH_{2n-2}
Alcohols	$C_nH_{2n+1}OH$
Carboxylic acid	$C_nH_{2n+1}COOH$

* C- Number table :-

1- Meth
2- Eth
3- Prop
4- But
5- Pen
6- Hex
7 Hep
8- Oct
9 - Non
10- Dec

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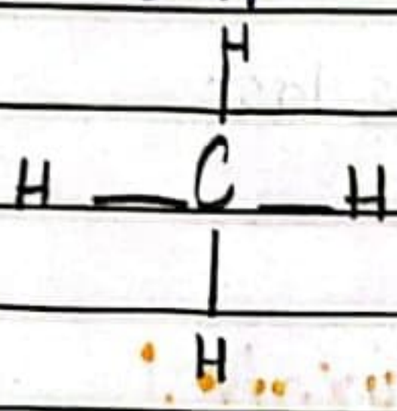
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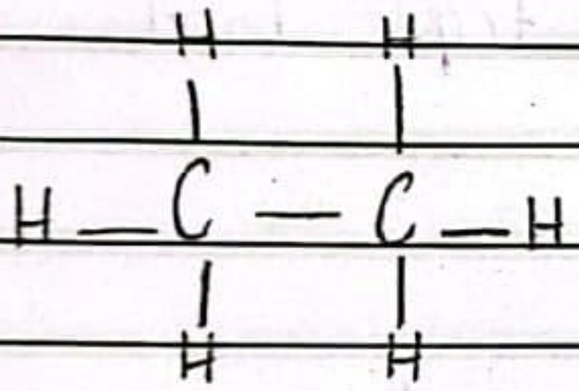
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Alkanes (Saturated) : (General formula : C_nH_{2n+2})

- Methane
CH_4



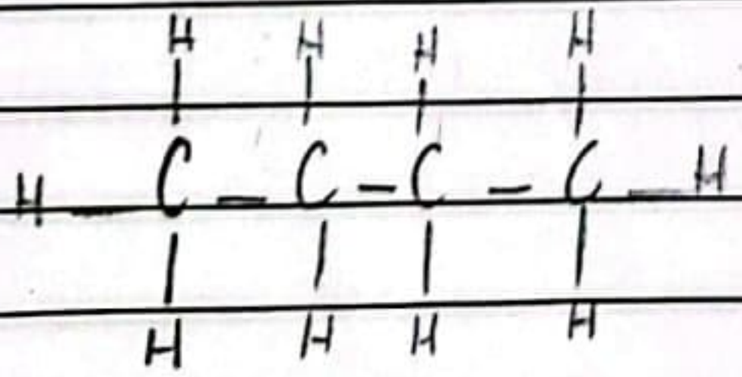
- Ethane (CH_3-CH_3)
C_2H_6
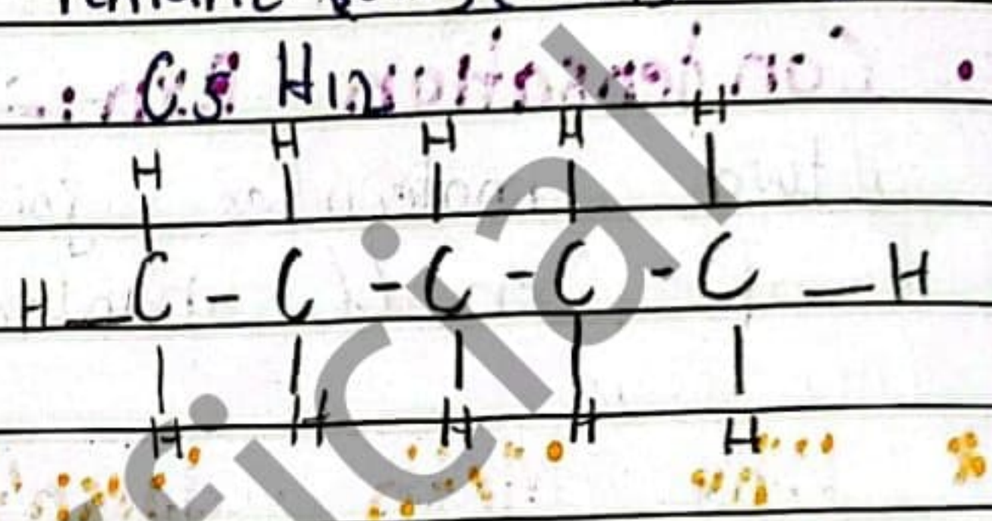


- Propane ($CH_3-CH_2-CH_3$)
C_3H_8



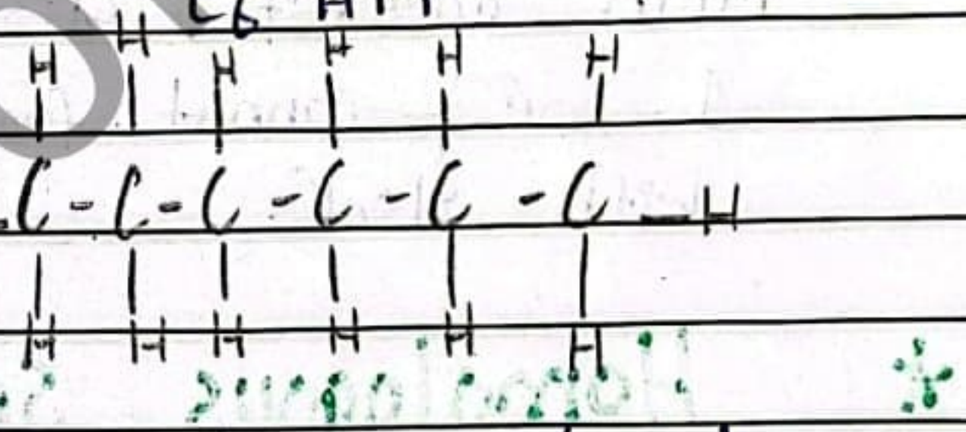
- Butane ($CH_3-CH_2-CH_2-CH_3$)
C_4H_{10}



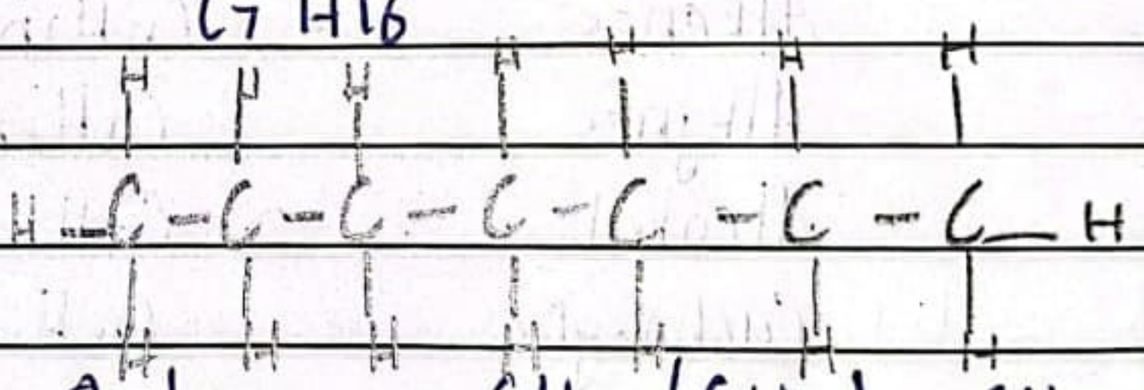
- Pentane $CH_3(CH_2)_3CH_3$
C_5H_{12}



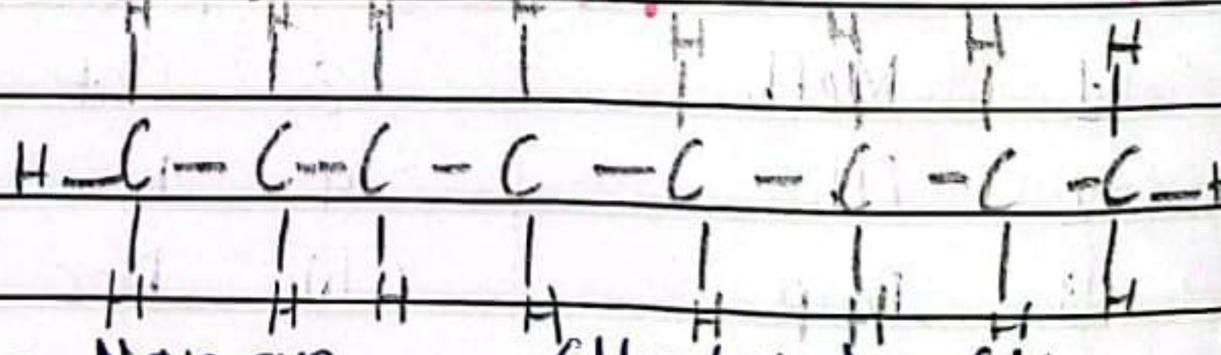
- Hexane $CH_3(CH_2)_4CH_3$
C_6H_{14}



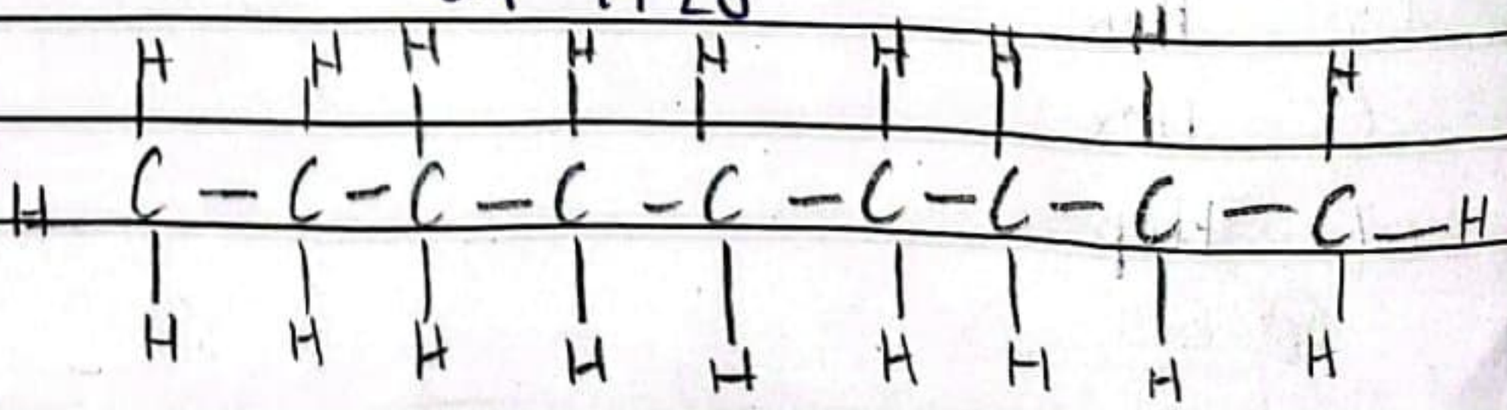
- Heptane $CH_3(CH_2)_5CH_3$
C_7H_{16}



- Octane $CH_3(CH_2)_6CH_3$
C_8H_{18}



- Nonan $CH_3(CH_2)_7CH_3$
C_9H_{20}



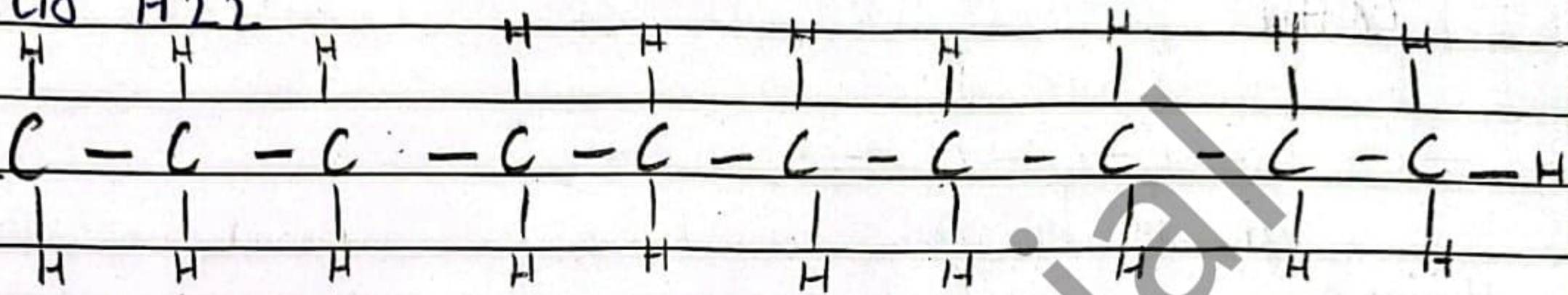
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10- Decane $CH_3(CH_2)_8CH_3$

$C_{10}H_{22}$



Alkenes (unsaturated) (General formula C_nH_{2n})

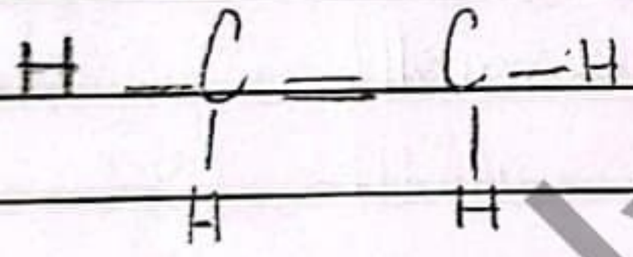
- Ethene

C_nH_{2n}

$n = 2$

$C_2H_{2(2)}$

C_2H_4



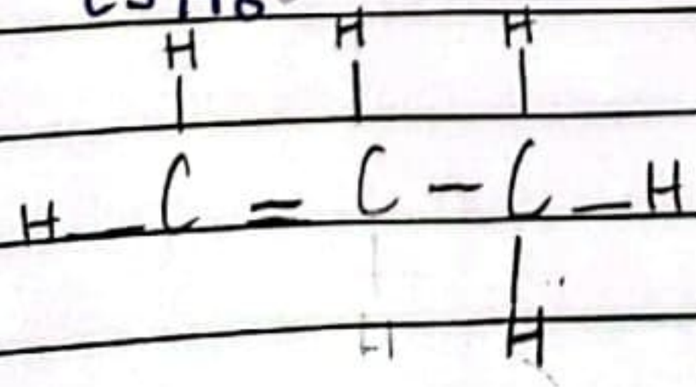
- Propene

C_nH_{2n}

$n_2 3$

$C_3H_{2(3)}$

C_3H_6



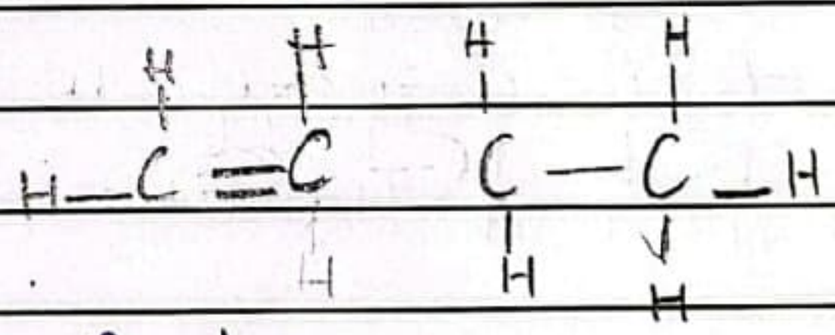
- Butene

C_nH_{2n}

$n = 4$

$C_4H_{2(4)}$

C_4H_8



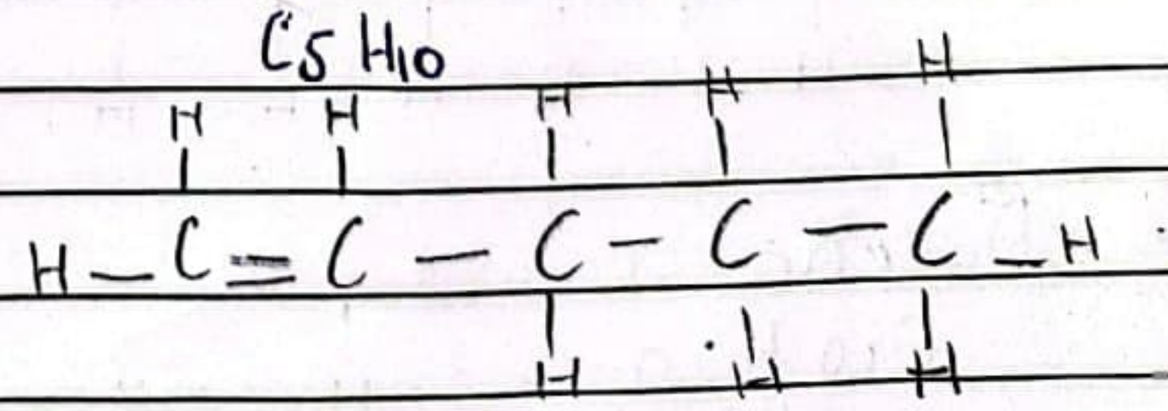
- Pentene

C_nH_{2n}

$C = 5$

$C_5H_{2(5)}$

C_5H_{10}



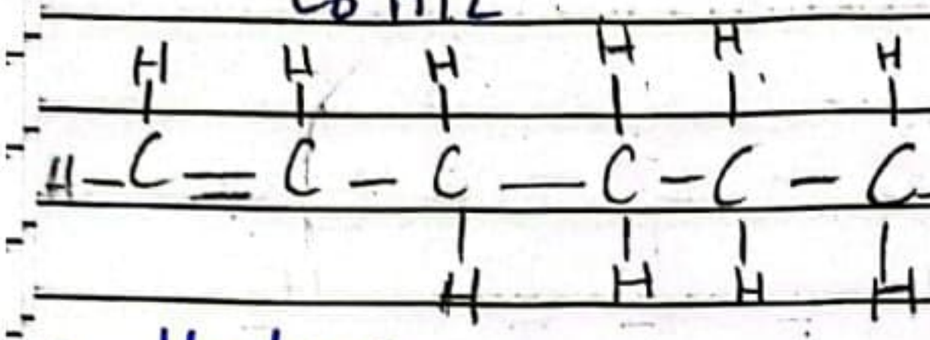
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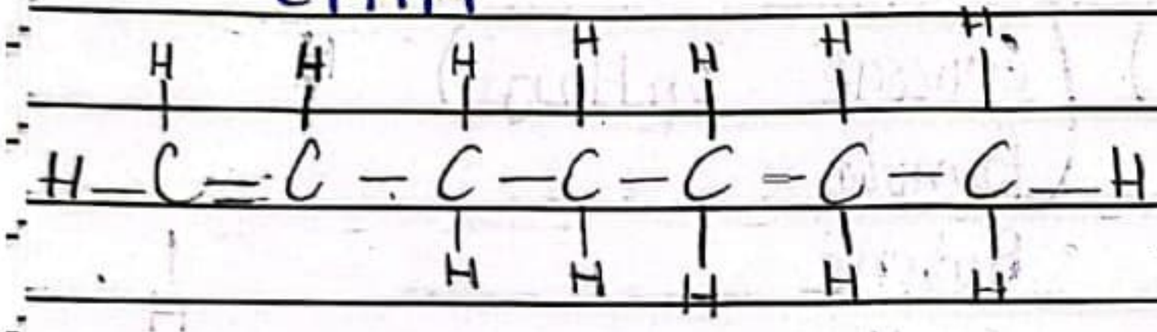
- Hexene

C_6H_{12}



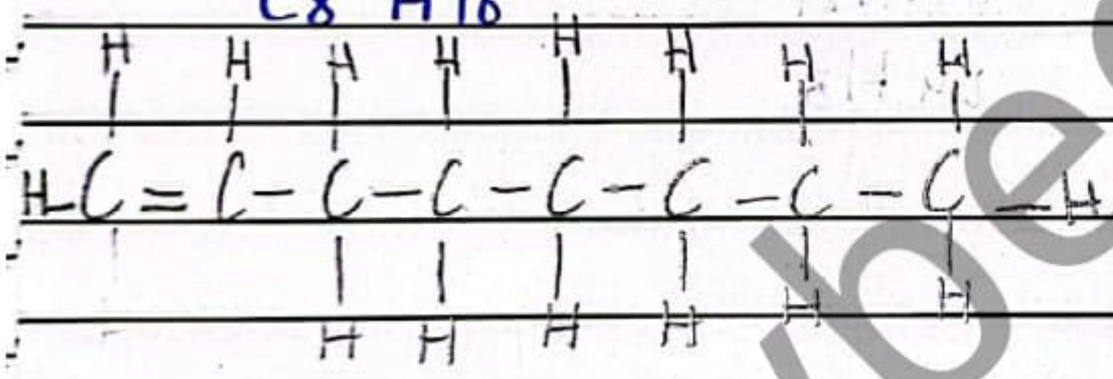
- Heptene

C_7H_{14}



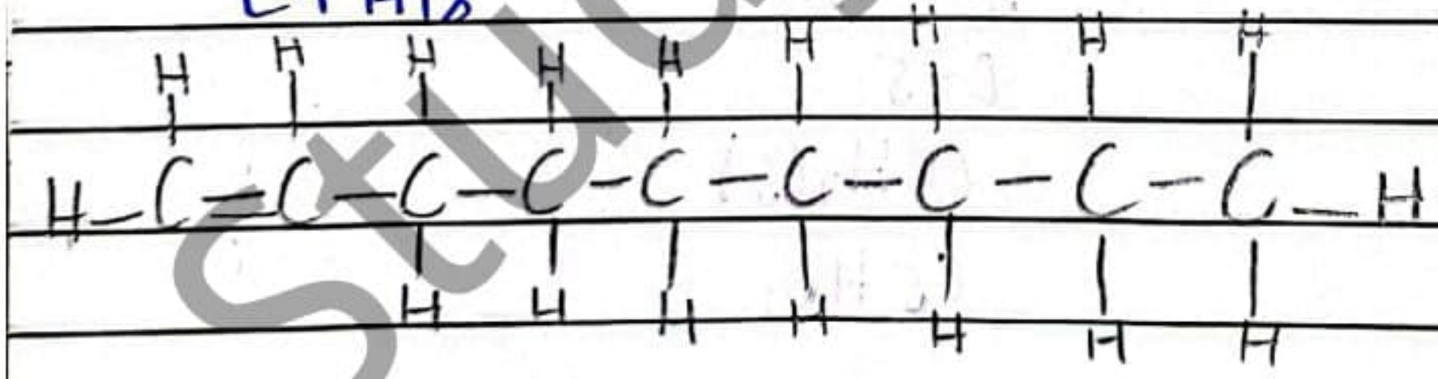
- Octene

C_8H_{16}



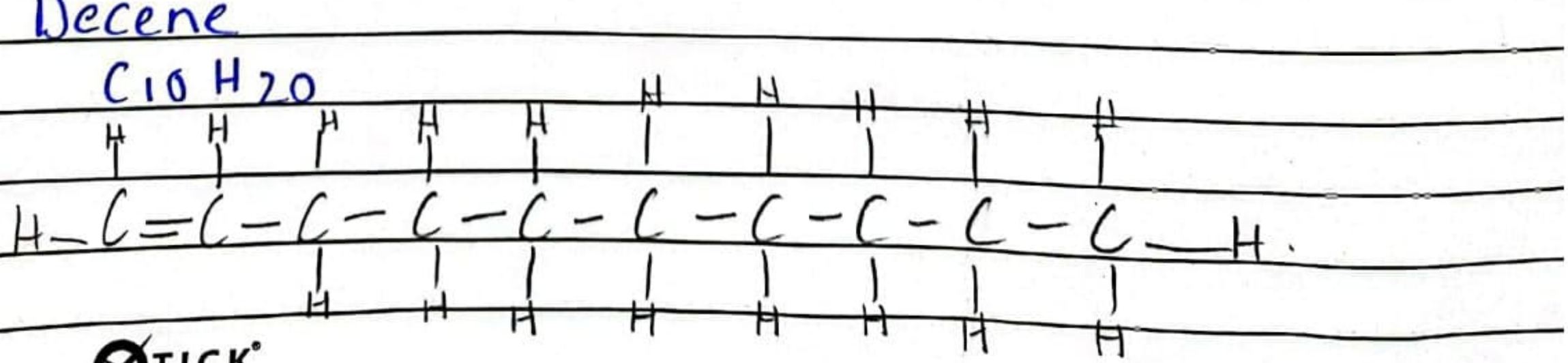
- Nonene

C_9H_{18}



- Decene

$C_{10}H_{20}$



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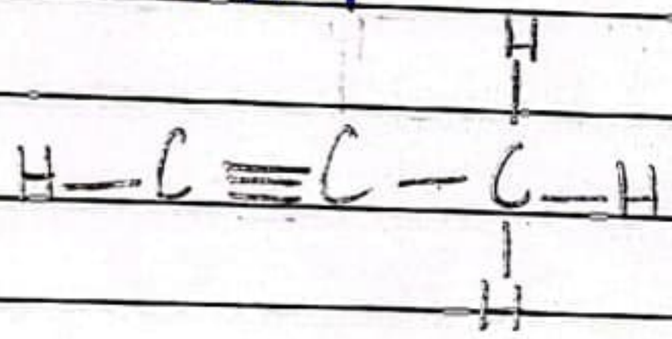
Alkynes (Unsaturated)

[General formula (C_nH_{2n-2})]

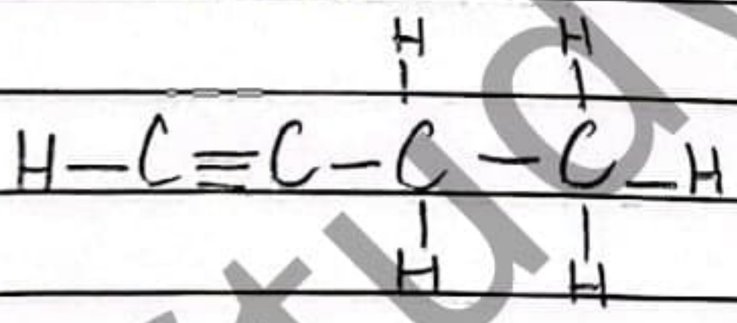
- Ethyne
  $C_2H_{2(2)-2}$
  C_2H_{4-2}
  C_2H_2



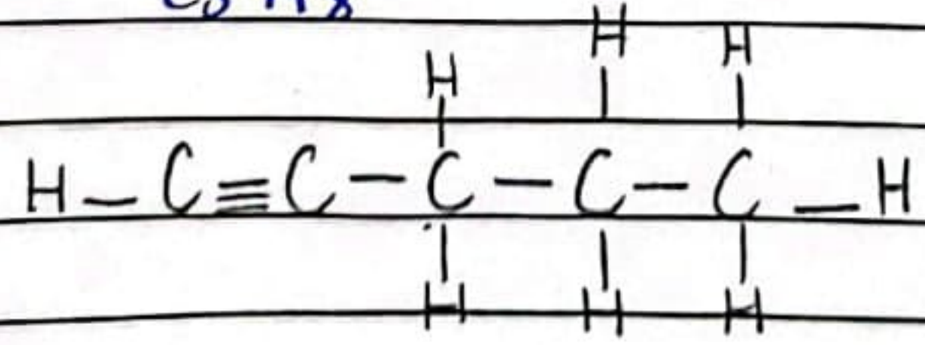
- Propyne
  C_3H_4



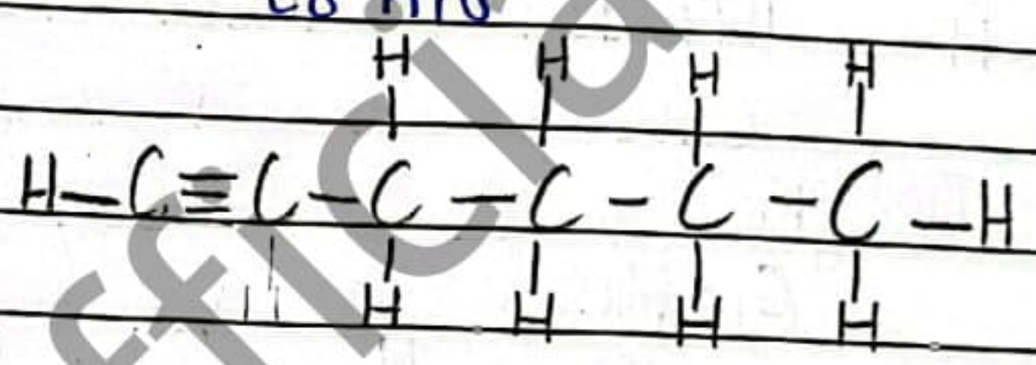
- Butyne
  C_4H_6



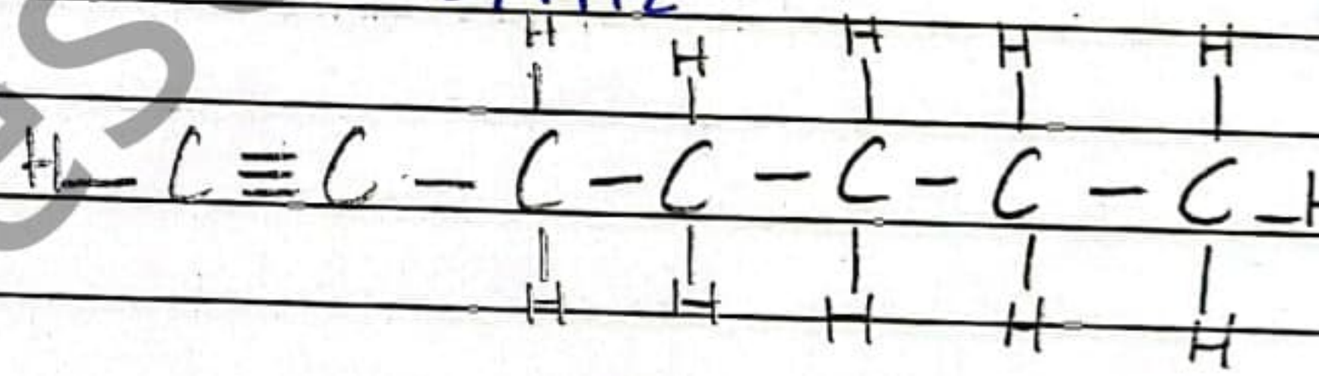
- Pentyne
  C_5H_8



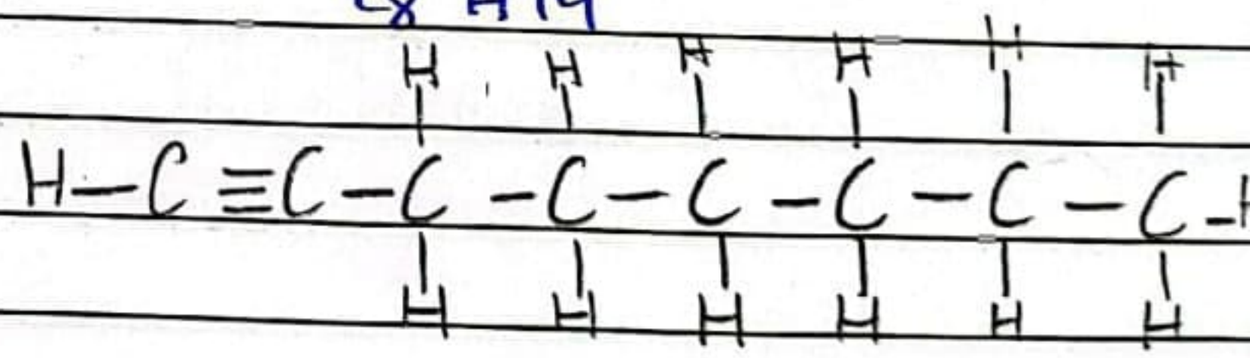
- Hexyne
  C_6H_{10}



- Heptyne
  C_7H_{12}



- Octyne
  C_8H_{14}



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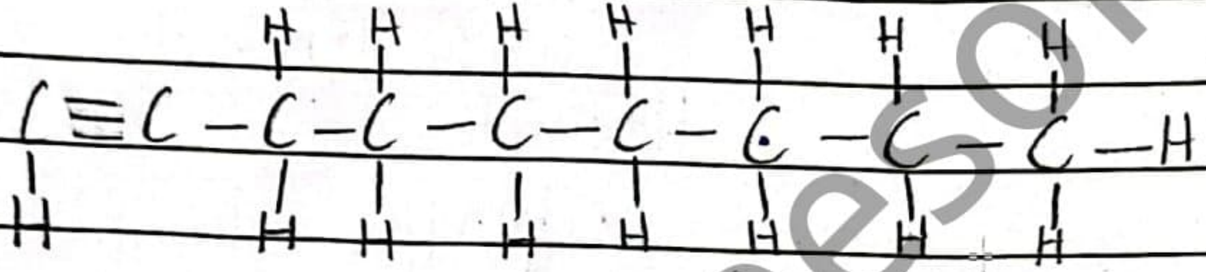
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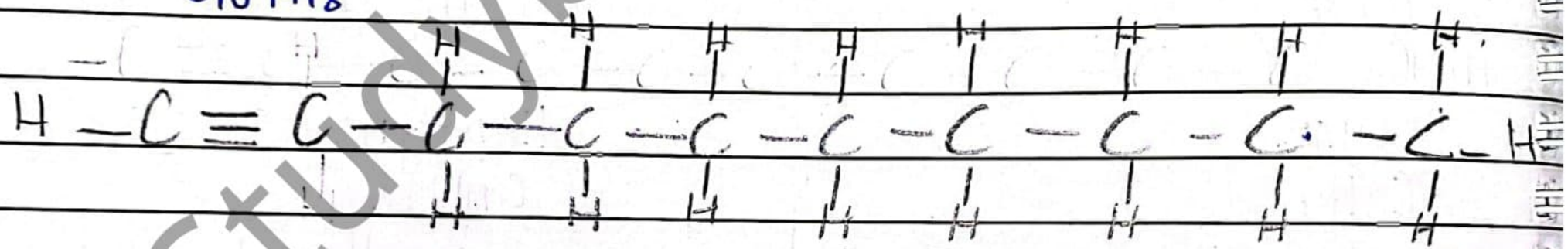
- Nonyne

C_9H_{16}



- Decyne

$C_{10}H_{18}$



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• Types of formulas :-

Molecular Formula :-

Shows the actual no. of each atom in a molecule.

e.g :-

Ethane = C_2H_6

General Formula

Represents any member of a homologous series using n. (number of carbon atoms).

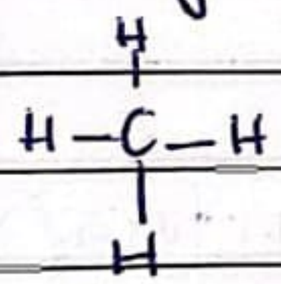
e.g :-

Alkanes = C_nH_{2n+2}

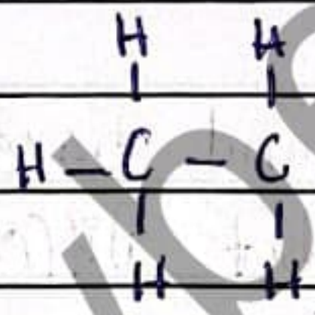
Structural formula

Describes the arrangement of atoms in a molecule.

e.g :-



Methane



Ethane.

Condensed structural form

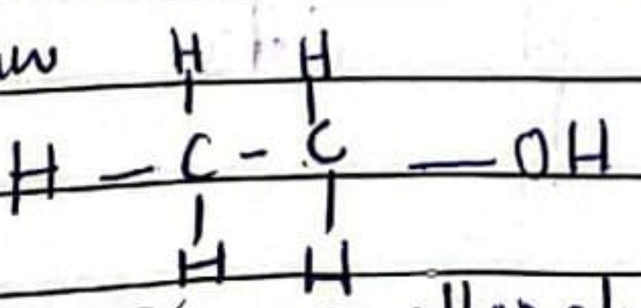
A simplified version of structural formula where we don't draw all the lines and bonds - instead we write each carbon with its hydrogen as a group in sequence.

e.g :-

CH_4 → Methane , CH_3CH_3 → Ethane.

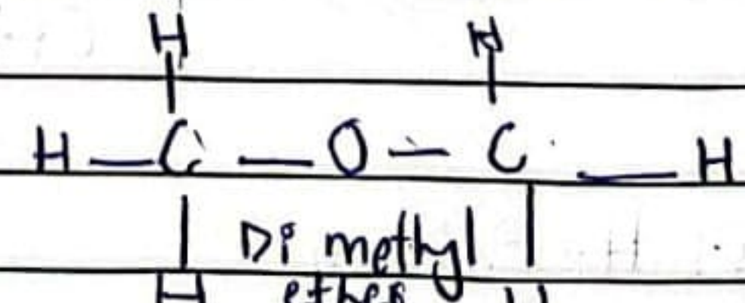
Structural vs Condensed Importance.

	Structural	Condensed	Importance.
Bonds shown	Every bond drawn	No bonds drawn	In Ethanol - O is attached to C, H
How it looks	Full diagram with lines (bonds).	Written as groups	Dimethyl ether - O is bonded to 2 C - atoms.
Time to draw	Takes longer	Quick n simple.	Same Molecular, totally diff structure, diff properties.



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ethanol



Dimethyl ether

Molecular formula = C_2H_6O

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Chemical Diversity

• Catenation :-

The self-linking ability of carbon to bond with other carbon atoms to form long chains, branched chains and rings.

→ Carbon has 4 valence $\bar{e}$ → can form 4 bonds.

→ No other element can compete carbon in this.

→ Carbon chains can contain thousand of carbon atoms.

→ Silicon can also catenate but forms only short chains.

• Isomerism :-

When two or more compound have the same molecular formula but different structural arrangements → structural isomers.

this phenomenon is called isomerism.

Structural isomerism :-

Straight chain - n

Carbons are connected in 1 continuous straight line.

e.g :-

n - Butane (C_4H_{10}) :-

$CH_3 - CH_2 - CH_2 - CH_3$

CH_4, C_2H_6, C_3H_8 → Non isomers

Branched chain - iso

Carbons have a branch / side chain coming of the main chain.

e.g.

Iso - butane (C_4H_{10})

$CH_3 - CH(CH_3) - CH_3$

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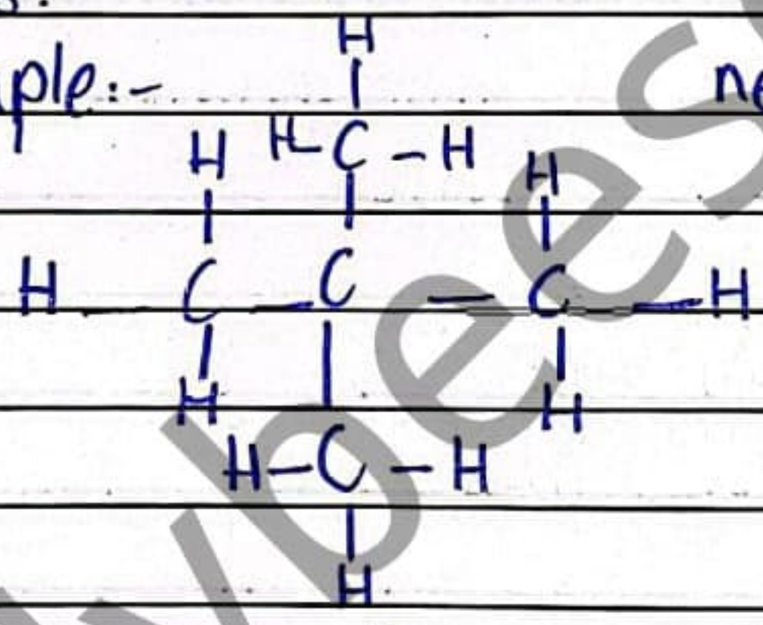
⟹ Chain isomerism begins with butane (C_4H_{10}) because compounds with fewer than four carbon atoms cannot be arranged in more than 1 way.

- **Neo-Chain:-**

→ highly branched chain (structure) with a carbon attached to four other carbons (a quaternary carbon).

→ Middle carbon is connected to four carbon groups.

Example:- neo-pentane ($C_5 10_{12}$)



- Remove 1 H from an Alkane → Alkyl group.

Alkane		Alkyl.	
Methane	CH_4	CH_3-	Methyl
Ethane	C_2H_6	C_2H_5-	Ethyl
Propane	C_3H_8	C_3H_7-	Propyl
Butane	C_4H_{10}	C_4H_9-	Butyl

- Remove H_2 from an Alkane → Alkene
- Remove another H_2 from an Alkene → Alkyne.

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Classification

Organic compounds are broadly classified into 2 main group based on their carbon skeleton :

→ Open chain or Acyclic compounds :-

These compounds catain an open chain of carbon atoms linked in a linear or branched pattern. They do not form rings.

For example :-

Straight chain

$CH_3 - CH_2 - CH_3$

propane

$CH_3 - CH_2 - CH_2 - CH_3$

Butane

Branched chain

$CH_3 - \underset{\displaystyle CH_3}{\underset{|}{CH}} - CH_3$

iso- butane

$CH_3 - CH_2 - \underset{\displaystyle CH_3}{\underset{|}{CH}} - CH_3$

iso - pentane.

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→ Closed chain or Cyclic Compounds :-

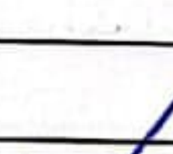
These are organic compounds that contain 1 or more rings of atoms.

Homocyclic Compounds
(Carbocyclic)

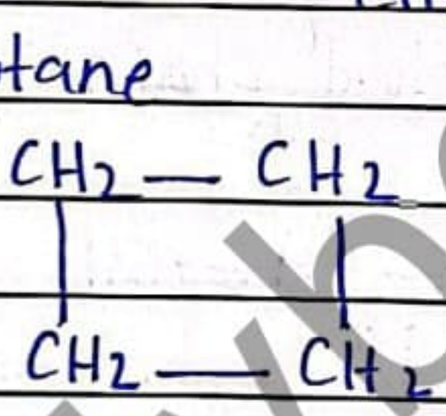
The main ring consist of only carbon atoms.

e.g.:-

Cyclopropane



Cyclobutane



Heterocyclic compou

The main ring contain carbon atoms along with an atom of another element.
(like nitrogen, O2, Sulph

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